

Hydrogen

1.

(1) Water electrolysis using renewable electricity, such as electricity generated from solar or wind power.

(2) Photoelectrochemical water splitting, in which sunlight is directly used to split water into hydrogen and oxygen.

(3) Biological or thermochemical hydrogen production using renewable biomass or solar energy: hydrogen production by photosynthetic microorganisms or thermochemical water-splitting cycles driven by solar heat.

利用可再生能源（太阳能、风能）发电来驱动电解水、利用太阳光直接分解水的“光化学分解”、利用高温热力循环分解水的“热化学分解”。

2.

Hydrogen is considered an energy carrier rather than a primary energy source because molecular hydrogen is not normally available in a readily usable form in nature. Energy must first be supplied to produce hydrogen.

Therefore, hydrogen stores and transports energy that has been supplied from another energy source. Renewable electricity can be converted into chemical energy stored in hydrogen through electrolysis. The hydrogen can later be converted back into electricity, heat, or mechanical work.

This differs from fossil fuels. Fossil fuels (coal, oil, and natural gas) are primary energy resources because they contain stored chemical energy that can be extracted directly from naturally occurring deposits. But hydrogen must generally be produced before it can be used as a fuel or energy carrier.

As is shown in the textbook, hydrogen is an energy carrier, it does not exist as a gas on Earth in a form that can simply be extracted for use; it has to be produced.

氢在地球上不能像煤或石油那样直接开采，必须通过电解、重整等过程从水或碳氢化合物中制造出来。化石燃料是直接取自自然的一次能源。氢作为载体可以与风能、太阳能等可再生且不稳定的能源配合，起到储存和调节的作用。

3.

Hydrogen can be stored using physical and material-based methods.

Physical-based storage:

(1) Compressed gas: Stored at high pressure (up to 800 atm) in tanks, often made of carbon fiber reinforced polymer (CFRP).

(2) Cryogenic storage: Liquefied at very low temperatures (below its boiling point of 20 K) in double-walled insulated tanks.

(3) Cryo-compressed: A combination of compression and cryogenic cooling.

Material-based storage:

(1)Chemical storage: Hydrogen is bound in chemicals like Ammonia (NH₃), Metal Hydrides (MH), or Liquid Organic Hydrogen Carriers (LOHC).

(2)Adsorbents: Stored on high-surface-area materials like metal-organic frameworks (MOFs).

Transportation,:Hydrogen can be moved as a pressurized gas via pipelines, as a cryogenic liquid in special tankers, or transported in a chemical precursor form. The choice depends on distance, cost, and safety.

储存分为物理类（加压、液化）和材料类（化学氢化物、金属氢化物、吸附材料等），可以用管道、高压气罐车、液氢槽车，或通过化学载体（如氨）进行运输。

4.

The minimum theoretical energy consumption for water electrolysis is equivalent to the change in Gibbs free energy (ΔG) for the water-splitting reaction. This is the minimum electrical energy required to drive the reaction. At standard conditions (25° C, 1 atm), the change in Gibbs free energy is approximately 237 kJ/mol. This corresponds to a theoretical minimum voltage of about 1.23 V (calculated from $\Delta G = -nFE$).

电解水是一个非自发反应，要驱动一个非自发反应，必须提供至少等于其吉布斯自由能变化的能量。这个能量对应于电解槽所需的最小理论电压（1.23 V），否则反应无法发生。

5.

The hydrogen economy is a future energy-system concept in which hydrogen plays a major role as an energy carrier, potentially replacing a significant part of the role currently played by fossil fuels. Renewable electricity can be used to produce hydrogen, which can then be stored, transported, and converted into electricity, heat, or mechanical work.

Technological and infrastructural bottlenecks :

(1)Efficient, low-cost, and clean methods to produce hydrogen (especially from renewable sources) need to be developed and scaled up.

(2)Hydrogen has an extremely low volumetric energy density,so safe, compact, and cost-effective onboard and stationary storage systems are required,

(3)A widespread network for hydrogen transportation and distribution (pipelines, refueling stations) is currently lacking and must be built.

(4)The cost and durability of fuel cells and electrolyzers need to be improved significantly to be competitive with conventional technologies.

氢的密度低、储存难、基础设施不完善、燃料电池成本高等都是需要解决的关键瓶颈。

6.

The reaction: $2H_2 + O_2 \rightarrow 2H_2O$,so we can see that hydrogen and oxygen are

combined in a 2:1 volume ratio, so the stoichiometric ratio is 66.7% H_2 and 33.3% O_2 . The broader range of compositions that can actually form an explosive mixture depends on temperature, pressure, ignition conditions, and the surrounding gas.

Batteries

1.

a. Gibbs free energy is a thermodynamic quantity that represents the maximum amount of non-expansion work that can be obtained from a system at constant temperature and pressure. It is defined by

$$G = H - T_{ref}S = U + p_{ref}V - T_{ref}S$$

The Gibbs energy is by definition related to enthalpy H , entropy S and temperature T . The change in Gibbs free energy ΔG gives a measure of the spontaneity

of a reaction. If $\Delta G < 0$, a reaction will occur spontaneously. On the other hand, if $\Delta G > 0$, the reaction is not spontaneous. If $\Delta G = 0$ the system is at equilibrium. According to thermodynamics, the maximum amount of chemical energy of a system that in a given situation can be converted into a high-quality energy form like electricity is given by the free energy G also called the Gibbs energy.

吉布斯自由能是热力学中用来衡量系统在恒温恒压下能对外做的最大有用功的物理量,吉布斯自由能的变化量决定了反应的自发性。

b. The basic unit in an electrochemical device is the electrochemical cell. In electrolytic cells, electric energy is used to bring forward chemical reactions.

电化学装置的基本单元是电化学电池,电池在放电时作为原电池工作,而在充电时则作为电解池运行,用电能来驱动化学反应。

c. A galvanic cell is an electrochemical cell in which a spontaneous chemical reaction produces electrical energy. In a voltaic or galvanic cell, chemical reactions are used to produce electrical energy.

原电池是依靠自发的化学反应直接产生电能的装置。在伏打或原电池中,化学反应被用来产生电能。

d. The battery capacity Q is the product of the current and time, i.e., $Q = I \cdot t$ and the unit is ampere hour (Ah) or milliampere hour (mAh).

电池容量 Q 是电流与时间的乘积,即 $Q = I \cdot t$, 单位为安时 (Ah) 或毫安时 (mAh)。

e. The product of the battery voltage and the rated ampere hours is called the Watt hours of the battery and is calculated as Watt Hours (Wh) = Battery Voltage * Ah ($V \cdot I \cdot t$).

电池电压与额定安时数的乘积称为电池的瓦时, 计算公式为: 瓦时 (Wh) = 电池电压 $\times$ Ah ($V \cdot I \cdot t$)。

f. The energy density is defined as the amount of energy stored per unit volume, i.e.,

Wh/m^3 . 能量密度指单位体积内存储的能量。

g. The state of charge is defined as the ratio of the available capacity $Q(t)$ and the maximum possible charge that can be stored in a battery, i.e., the nominal capacity

Q_n .

$$SOC(t) = \frac{Q(t)}{Q_n}$$

SOC 指电池当前剩余容量与额定容量的比值。

h. The depth of discharge (DOD) is used to describe how deeply the battery has been discharged. In most cases it is related to SOC by $DOD = 100 - SOC(100\%)$.

DoD 是放电深度，指电池已经放出的电量占总容量的百分比。

i. The electrochemical cell is composed of two so-called half-cells, each composed of an electrode dipped in a solution of an electrolyte. These half-cells are designed to contain the oxidation half reaction and reduction half-reaction separately.

半电池是构成电池的基本单元之一，由一个电极和对应的电解质溶液组成，只包含氧化或还原反应中的一半。

j. The reactions at the electrodes are classified as an oxidation reaction, in which electrons are removed or released from a reactant, and as a reduction reaction, in which electrons are consumed by or added to another reactant. The reduction oxidation processes are summarized as redox reactions.

电极上的反应分为氧化反应和还原反应。氧化反应是指从反应物中移除或释放电子，而还原反应则是指电子被另一个反应物消耗或加入到其上。这些氧化还原过程统称为氧化还原反应。

k. The state of health is an indicator of the remaining battery capacity compared to the capacity at BOL. The aging processes, which lower the SOH, start as soon as the cells are assembled. SOH is not a physical quantity but it depends on parameters like number of charge-discharge cycles, capacity and power fade, and internal resistance. The aging process of a battery is complex and depends on the involved mechanisms. The aging depends on the vehicle usage, battery and cell design. Temperature, SOC range, energy and power conditions and time are essential parameters.

健康状态是衡量电池剩余容量与初始容量（BOL）相比的指标。电池老化过程从组装电池开始就已启动，会逐渐降低健康状态（SOH）。SOH 并非物理量，而是取决于充放电循环次数、容量衰减、功率衰减以及内阻等参数。电池的衰老过程复杂，受多种机制影响，同时与车辆使用情况、电池及电芯设计密切相关。温度、SOC 范围、能量和功率条件以及时间都是关键影响因素。

l. Degradation refers to the irreversible loss of battery performance (capacity and power) over time due to aging processes. It is connected with the growth of the SEI (solid-electrolyte interface)-layer on the anode, loss of cathode material, loss of conductive path, plating of metallic lithium and electrolyte oxidation. If electrons would be transferred into the electrolyte, reduction of the number of Li ions being converted to metallic Li will appear.

退化是指电池因老化过程而随时间推移不可逆地丧失性能（容量和功率）。这与阳极上固态电解质界面（SEI）层的增厚、阴极材料损耗、导电通道损失、金属

锂析出以及电解液氧化有关。锂电池中，如果电子转移到电解液中，将导致转化为金属锂的锂离子数量减少。这是锂离子电池的一种退化机制。

m. There are a number of ways in which cooling of a battery pack can be done. The system for this is generally called the battery thermal management system (BTMS). The tasks of the BTMS are to maintain the operating temperature within 5-30°C, maintain the temperature difference between any cells in the pack to be less than 5 °C, prevent thermal runaway, maximize the useful energy from cells and pack, and use a small amount of energy for operation. The types of BTMS are classified as active or passive systems. In a passive cooling system, the heat is dissipated through natural convection and thermal radiation. These have no moving parts and are sufficient if a large space is available or if low performance (i.e., low charge and discharge rates) is required. The thermal inertia of the system is important and sometimes this is increased by adding more thermal mass. Active cooling systems rely on a cooling fluid, which is forced through the battery by a fan (air cooling) or pump (liquid cooling).

电池组的冷却方式有多种，其系统通常称为电池热管理系统（BTMS）。BTMS 的主要任务包括：将工作温度维持在 5-30°C 之间，保持电池组内各电芯之间的温差小于 5°C，防止热失控，最大限度地利用电芯和电池组的可用能量，并以较低能耗运行。BTMS 可分为主动式和被动式两种类型。在被动式冷却系统中，热量通过自然对流和热辐射散失，该系统无运动部件，适用于空间较大或要求低性能（如低充放电速率）的情况。系统的热惯性较大，有时可通过增加热质量来增强这一特性。而主动式冷却系统则依赖冷却流体，由风扇（风冷）或泵（液冷）驱动流经电池进行冷却。

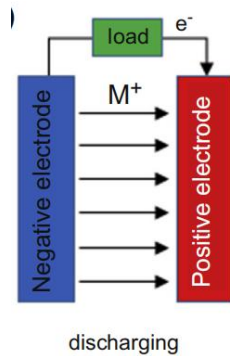
n. To operate an electric vehicle, the battery management system (BMS) should be able to inform about the power and energy status of the battery. The battery management system (BMS) keeps the battery at about 50% charge to allow absorbing energy on wind gusts and delivering on high load demands. Modern BMS can switch between charging and discharging within less than a second. This is favorable for stabilizing the voltage on the transmission lines (frequency regulation).

电池管理系统（BMS）能够负责监控和管理电池包的运行状态。现代 BMS 可以在不到一秒的时间内在充电和放电之间切换。这有利于稳定输电线路上的电压。

o. C-rate is a measure of the discharge or charge rate relative to the maximum cell capacity. 1C rate is a discharge current discharging the cell within 1 h.

C 倍率是描述电池充放电电流大小相对于其容量的标准化参数。1C 率表示 1 小时放完电池全部电量的电流。

2.

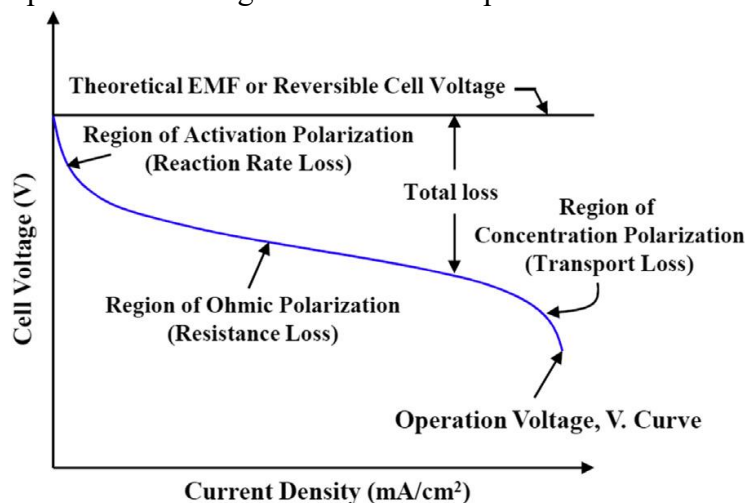


The electrode subject to oxidation is called the anode or the negative electrode. During the discharge process positive ions (cations) and electrons are produced. The cations move through the electrolyte to the positive electrode while the electrons pass the external circuit and then reach the positive electrode. At the positive electrode then a reduction process occurs with the incoming cations and electrons.

发生氧化的电极称为阳极或负极。在放电过程中会产生正离子（阳离子）和电子。阳离子通过电解质移动到阳极，而电子则穿过外部电路，最终到达阳极。在阳极处，进入的阳离子与电子发生还原反应。

3.

When a battery operates under load, the cell voltage is lower than its equilibrium or open-circuit voltage because several polarization losses occur.



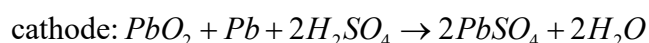
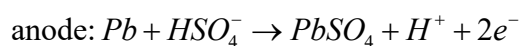
(1) Activation losses are dominant at low current densities, caused by the kinetics of the electrochemical reactions. Described by the Tafel equation.

(2) Ohmic polarization at intermediate current densities, caused by the resistance to ion flow in the electrolyte and electron flow in the electrodes. It increases linearly with current.

(3) Mass transport loss or concentration loss at high current densities, caused by mass transport limitations (reactants cannot reach the reaction sites fast enough).

Characterized by a limiting current density.

4.



5.

$$E_{cell} = E_{cell}^0 - \frac{RT}{n_e F} \ln Q_r$$

E is the cell potential at some instant of time, E^0 is the cell potential when the reaction is at the standard state condition, R is the ideal gas constant (8.314 J/mol, K), T is the temperature in degree Kelvin (K), n_e is the number of moles of electrons transferred in the reaction, F is the Faraday constant (the charge of a mole of electrons), Q_r is the reaction quotient introduced above.

6.

Nernst equation is an important equation in electrochemistry and it relates the reduction potential of an electrochemical reaction (half-cell or full cell reaction) to the standard electrode potential, temperature and activities (these latter ones are often approximated by concentrations) of the chemical species involved in the reduction and oxidation processes. The Nernst equation can be derived from the standard changes in Gibbs free energy associated with an electrochemical transformation. The full cell voltage is found by subtracting the anode potential from the cathode potential ($E_{cell} = E_{cathode} - E_{anode}$). Under open-circuit conditions (no current flowing), the cell voltage is the open-circuit voltage (OCV), which is given by the Nernst equation. It shows how the voltage changes with the state of charge (concentration of reactants and products) and temperature.

先分别计算两个半电池的电位，然后相减得到整个电池的电压。当电流为零时，此电压即为开路电压。

7.

$$i = i_0 \left[e^{\frac{\alpha n_e F}{RT} \eta} - e^{-\frac{(1-\alpha) n_e F}{RT} \eta} \right]$$

n_e is the number of moles of electrons involved in the electrochemical reaction .

α is the fraction of the electrode potential involved in the reaction.

i_0 : At an equilibrium state, the cathodic and anodic currents have the same magnitude and this value is called the exchange current density and is labeled as i_0 .

8.

The first law states that the amount of chemical change being produced by a current at an electrode-electrolyte interface is proportional to the quantity of electricity used. The second one tells that the amounts of chemical changes produced by the same quantity of electricity in different substances are proportional to their equivalent weights.

$$m_i = \frac{\xi Q M}{n_e F}$$

9.

EV:The simplest type among the electric vehicles as the battery is the single source of energy.

PHEV:Plug-in hybrid electric vehicles,the battery can be charged externally and the all-electric driving range will be longer than for the strong HEVs and the overall driving range will be longer than for the BEVs. The capacity of the battery is important for the all-electric driving range. The ability to recharge the battery externally by grid electricity, regenerative brake energy or by the internal combustion engine creates opportunities to optimize the energy consumption and accordingly the selection of battery.

HEV:hybrid electric vehicle,a vehicle having two or more types of power sources and at least one is an electric one. Brake energy is regenerated and stored in the battery and this is an important feature of HEVs concerning the total energy efficiency. The battery of an HEV can only be charged by recuperating the brake energy or utilizing the internal combustion engine, i.e., no external charging procedure is required.

纯电动车：电动汽车中能量来源最简单，电池是唯一的能源。

插电式混合动力汽车。

（PHEV）：电池可外部充电，全电动续航里程比强型混合动力车更长，整体续航能力也优于纯电动车。电池容量对全电动续航里程至关重要。通过电网电力、再生制动能量或内燃机为电池充电的能力，有助于优化能耗，从而影响电池的选择。

混合动力汽车（HEV）：具有两种或以上动力源的车辆，其中至少一种为电动动力。制动能量被回收并储存在电池中，这是提高整车能效的重要特点。HEV 的电池只能通过回收制动能量或利用内燃机进行充电，因此无需外部充电过程。

10.

A Li-ion battery operates on the principle of lithium-ion intercalation. During discharge, lithium ions (Li^+) leave the anode (typically graphite) and move through the electrolyte to the cathode (typically a metal oxide like LiCoO_2). This flow of ions is accompanied by an equal flow of electrons through an external circuit, producing electrical power. During charging, the process is reversed. The electrolyte is a lithium salt dissolved in an organic solvent, and it is chemically inert, allowing only Li^+ ions to pass while blocking electrons.

锂离子电池的工作原理是锂离子嵌入。在放电过程中，锂离子（ Li^+ ）从阳极脱离，穿过电解质流向阴极。离子的流动伴随着电子通过外部电路的相应流动，从而产生电能。在充电过程中，这一过程则相反。电解液为溶于有机溶剂中的锂盐，仅允许 Li^+ 离子通过，同时阻止电子的流动。

11

$$\Delta G_{\text{cell}} = -n_e F E_{\text{cell}}^0$$

12.

The theoretical capacity of an electrode is related to the charge transferred in a

specific reaction given by Faraday's law.

Actual capacity: The real amount of charge a battery delivers under specific operating conditions (current, temperature, etc.). It is always less than the theoretical capacity.

Rated capacity: The nominal capacity specified by the manufacturer, typically defined under a specific discharge rate and temperature.

13.

In an electrochemical cell it is desirable that the Gibbs energy can be converted to electrical energy in a fuel cell as well as in a battery during discharge. However, as polarizations occur as a current passes through the cell, energy losses appear. These polarizations result in consumption of Gibbs energy, which is released as heat energy during the electrochemical reactions.

The generated heat has two parts, a reversible one and an irreversible one. The reversible one is calculated by the change in entropy ΔS and then $Q_{rev} = T\Delta S$. The

irreversible part may be calculated by the Ohmic heat $Q_{irrev} = RI^2$ or as

$$Q_{irrev} = I(E_{rev} - V_C)$$

14.

This phenomenon is caused by an over-charging or an over-discharging rate which produces more internal heat than the battery can dissipate. The generated heat which cannot be dissipated causes the cell temperature to rise. This temperature rise makes the chemical reactions within the cell to take place at a faster rate. Accordingly, the increased rate of the chemical reactions generates even more heat that cannot be dissipated.

There are several mechanisms or processes that can trigger the event of thermal runaway, namely over-charge, over-discharge, external short-circuit, over-heating, internal short-circuit caused by crash or dendrite formation or impurities caused by the manufacturing.

温度升高导致反应加速，反应加速产生更多热量，热量又进一步升高温度。触发因素包括过充、过放、短路、过热或内部缺陷。

15.

Air cooling: For applications of batteries where extreme performance or heavy duties are not required, air cooling can be employed. BTMS0 based on air cooling are generally low cost alternatives and less complicated than liquid cooling systems. However, with an air cooled system it is hard to maintain an even temperature distribution across the battery pack.

Liquid cooling: This cooling method is suitable for both pouch cells and rectangular prismatic cells. It occupies a small space and provides direct cooling to at least one face of a cell. Among the disadvantages are the high design cost and that leakage between the interfaces may occur if the sealing is not adequately designed.

Phase Change Material (PCM) cooling: The method of a cooling matrix is commonly employed for cylindrical cells. A hollow cooling shell is manufactured with holes to

contain the cells. The cooling liquid enters at one end of the jacket and exits at the diagonal end.

空气冷却适用于低性能需求；液体冷却适用于矩形棱柱电池；PCM 通常用于圆柱形电池

16.

Portable electronics: Highest priority on energy density and safety . They have a moderate cycle life because devices are often replaced.

Electric vehicles: High priority on energy density for driving range, power density for acceleration, and cycle life. Safety is also important due to the large size of the battery packs.

Grid-scale energy storage: The highest priority is cycle life to maximize the lifetime investment and cost. Energy and power density are less critical . Safety is also important.

17.

The end-of-life (EOL) is connected with the growth of the SEI (solid-electrolyte interface)-layer on the anode, loss of cathode material, loss of conductive path, plating of metallic lithium and electrolyte oxidation.

Extending methods:

(1) Maintain optimal temperature, avoid high temperatures that accelerate degradation and low temperatures that can cause lithium plating.

(2) Limit State of Charge (SOC) range, avoid full charges and deep discharges.

(3) Use an effective Battery Management System (BMS) to monitor and control the battery within safe operating limits.

Fuel Cells

1.

A fuel cell is a galvanic cell, it converts the chemical energy of a supplied fuel into electrical energy through a spontaneous electrochemical reaction.

2.

PEMFC:

principals :Hydrogen is oxidized at the anode, producing protons (H^+) and electrons. The protons travel through a polymer electrolyte membrane (Nafion), while the electrons travel through an external circuit. At the cathode, oxygen is reduced and combines with the protons and electrons to form water.

components:the anode, cathode and membrane. The electrode and membrane are often manufactured together and then called the membrane electrode assembly (MEA). The membrane is basically a hydrated solid which promotes the conduction of protons. Many different membrane types exist but the far most used is called Nafion being produced by DuPont. The Nafion layer is basically a carbon chain which has a fluorine atom layer attached to it. platinum catalyst, and gas diffusion layers (carbon fiber paper). It operates at low temperatures .

SOFC:

principals :Oxygen from the air is reduced at the cathode to form oxide ions (O^{2-}). These ions travel through a solid ceramic electrolyte. At the anode, the fuel (like hydrogen) reacts with the oxide ions to produce water and release electrons.

components:Anode (Ni-YSZ cermet), cathode (e.g., lanthanum manganite), solid oxide electrolyte (YSZ), and interconnects. It operates at high temperatures and can reform fuels internally.

3.The enthalpy of formation of the product in a chemical reaction is the difference in enthalpy of the product and the sum of enthalpy of all reactants. The reference state is defined by a temperature of 25 °C and a pressure of 0.1 MPa. At this state the enthalpy of the reactants is zero. The enthalpy of the product at the reference state is the given by the net heat transfer and this is termed the enthalpy of formation for the product.

化学反应中产物的生成焓是产物的焓与所有反应物焓之和的差值。参考状态定义为温度 25°C、压力 0.1 MPa。在此状态下，反应物的焓为零。产物在参考状态下的焓由净热量传递决定，这被称为产物的生成焓。

4.

The i_0 is the so-called exchange current density for a single electrode. It is related to the currents associated with the dynamic equilibrium existing at the electrode-electrolyte interfaces. At the interfaces a dynamic exchange of material between the electrode and the electrolyte occurs. Because ionic species are involved in the exchange process, anodic and cathodic currents are associated with it. At equilibrium, the anodic and cathodic currents are equal and are referred to as the exchange current density of the system. A high exchange current density indicates a fast reaction and a highly reversible system.

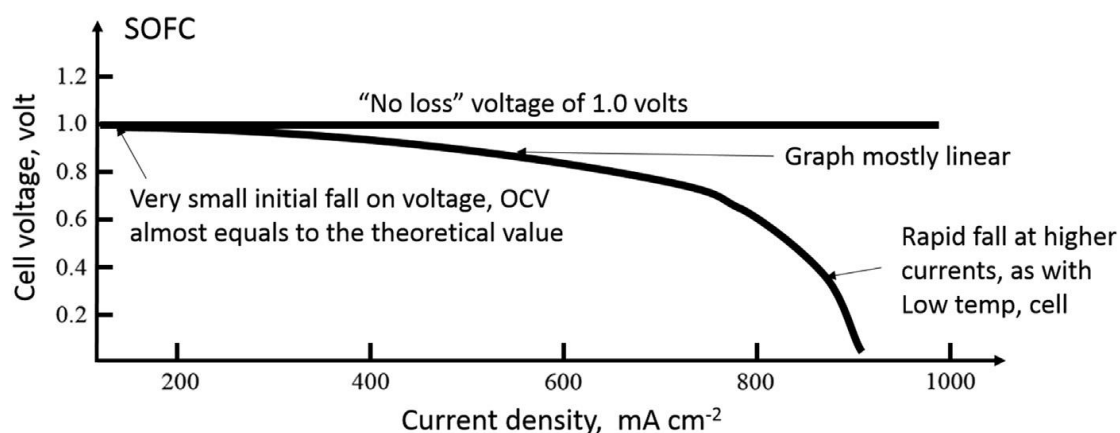
5.

Activation polarization region:the sluggish kinetics of the electrochemical reactions.

Ohmic polarization region: the resistance to charge transport (ions and electrons) within the cell components.

Concentration polarization region: mass transport limitations.

6.



The cell voltage that starts at a high open-circuit voltage (OCV). It has a very small initial drop at low current densities (less activation loss than a PEMFC), followed by a

mostly linear decrease in the middle region, which is dominated by ohmic losses. Finally, at high current densities, the voltage rapidly falls due to concentration (mass transport) losses.

The efficiency of a fuel cell :The efficiency of a fuel cell is defined as the ratio of the electrical energy output to the chemical energy input.

7.

$$\eta_{act} = a \cdot \log \frac{i}{i_0} = A \cdot \ln \frac{i}{i_0}$$

a and A are constants (Tafel slopes), and i_0 is the exchange current density.

8.

The first law of thermodynamics states that

$$du = q - p dv = T ds - p dv;$$

$$\text{and } dG = V dp - S dT;$$

If an isothermal process is considered, one has

$$dG = V dp;$$

The ideal gas states that

$$pV = nRT;$$

$$\text{so } dG = nRT \frac{dp}{p}$$

$$G = G^0 = nRT \ln \frac{p}{p_0}$$

$$g = g^0 + RT \ln \frac{p}{p_0}$$

9.

Anode: Porous carbon paper (gas diffusion layer) coated with a platinum catalyst.

Cathode: Porous carbon paper (gas diffusion layer) coated with a platinum catalyst.

Electrolyte: A solid polymer membrane, typically Nafion, which conducts protons when hydrated.

10.

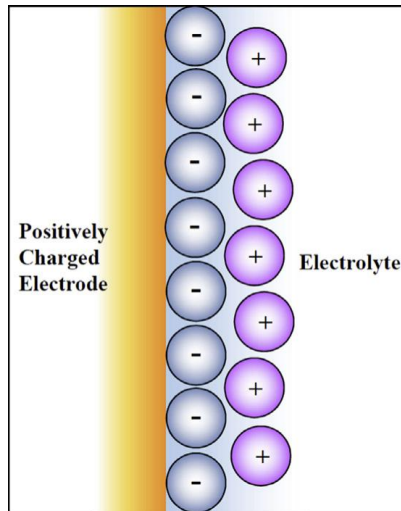
Anode: A Ni-YSZ cermet (a mixture of nickel metal and yttria-stabilized zirconia ceramic).

Cathode: A doped lanthanum manganite (e.g., LaMnO_3), a perovskite material.

Electrolyte: Yttria-stabilized zirconia (YSZ), a solid ceramic material that conducts oxide ions (O^{2-}).

11.

As a metal electrode is placed in an electrolyte, the charge of the metal will attract ions of opposite charge in the electrolyte. Then a layer of charge is formed both in the electrolyte and the metal. This layer is called the electrical or electrochemical double layer.



12.

$$\text{Oxygen consumption rate: } \dot{m}_{O_2} = \frac{P_{total}}{V_C} \frac{1}{n_e F} M_{O_2}$$

$$\text{Hydrogen consumption rate : } \dot{m}_{H_2} = \frac{P_{total}}{V_C} \frac{1}{n_e F} M_{H_2}$$

$$\text{Water production rate : } \dot{m}_{H_2O} = \frac{P_{total}}{V_C} \frac{1}{n_e F} M_{H_2O}$$

13.

Activation loss: The voltage loss due to the energy needed to drive the electrochemical reactions at the electrode-electrolyte interfaces.

Ohmic loss: the resistance to the flow of electrons and ions through the cell components (electrolyte, electrodes, interconnects).

Mass transport loss or concentration loss: the depletion of reactants at the reaction sites due to slow diffusion, especially at high current densities.

Internal current and crossover: The electrolyte is primarily aimed to transport ions but it may also have a small ability to conduct electrons. In fuel cells a small amount of the fuel may migrate through the electrolyte. However, these effects are not of great significance for the operating efficiency and are omitted in most approaches of analysis.

14.

Heat and mass transfer: The reactant gases flow through the gas flow channels and reactant gas species are transported from the gas flow channels and through the porous electrodes. Ions are transported through the membranes or electrolyte while electrons are transported through the electrodes and interconnects.

Charge and water transport: Electrons and ions are produced and consumed in two electrochemical reactions at the anode-electrolyte and cathode-electrolyte interfaces. The electrons are transported through the electrodes and interconnect to the external electric circuit. Ions are transported through the electrolyte from the electrode where it

is produced to the electrode where it is consumed. Ohmic voltage losses are caused by the resistances to the motion of ions through the electrolyte as well as electrons through the electrodes, interconnect materials, and contact interfaces. Ion transport is also important for the transport of water in PEM fuel cells.

15. Flooding of the electrolyte-electrode interface regions would block the pores of the electrode-gas diffusion layer and prevent the reactant gases to reach the reaction sites. This causes a loss due to concentration polarization or mass transfer loss.

The water flooding and mass transfer loss are more critical for PEMFCs because the diffusivity of oxygen in water at the cathode side is quite low. Water management is a key issue but also the polymer membrane (electrolyte) of PEMFCs needs to be hydrated to provide higher proton conductivity.

16.

Porosity: The fraction of the total volume of a porous medium that is occupied by the void (pore) space. It is a measure of how much space is available for fluid to flow.

Tortuosity: The tortuosity τ is defined as the ratio between the shortest pathway s through a

porous structure with side length L , i.e.,

$$\tau = \frac{s}{L}$$

Permeability: A measure of the ability of a porous medium to allow fluid to flow through it. It represents how easily a fluid can pass through the pores under a pressure gradient, as defined by Darcy's law.

$$17. u_{k,j} = -K \frac{k_{rk}}{\mu_k} (p_{k,j} - \rho_k g_j)$$

K is the permeability of the porous medium, μ_k the dynamic viscosity of

phase k , k_{rk} is the relative permeability of phase k and g_j is the gravity vector.

18. As the flow rate is increased, inertia forces become more significant and the linear relation does not hold and instead a non-linear relation is needed. The relation between the pressure gradient $p_{k,j}$ and velocity $u_{k,j}$ is then called the Forchheimer equation and for a one-dimensional case it is written as:

$$-\frac{\partial p}{\partial x} = aV + bV^2$$

a and b are constants.